

PIT: A Novel Toothbrush Providing Real-Time and Robust Plaque Indication During Brushing

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ABSTRACT

The dental plaque disclosing agent helps guide plaque removal by revealing plaque on the teeth. However, existing toothbrushes cannot provide visualization of the stained plaque beneath the toothpaste foam during brushing. To fill this gap, this paper proposes PIT and uses smartphone to provide real-time plaque visualization during brushing. PIT introduces novel hardware, including a micro camera, four green LEDs, and a mechanical structure, offering a stable camera view of bristle position and bristle rotation. To address the challenge of toothpaste foam obstruction, we derived an optical channel model that guided the design of the light source to enhance plaque visibility. Furthermore, we designed a deep neural network specifically for plaque segmentation under thick foam. Finally, the trained distilled student model was run on a smartphone and evaluated on both denture models and human subjects. The results show that PIT achieved an IoU (Intersection over Union) of 75.22% and a latency of 29 ms, with robust performance across various conditions. Evaluation by 10 participants revealed that PIT helped reduce plaque coverage to 5.6% within two minutes of brushing, significantly outperforming existing advanced toothbrushes.

CCS CONCEPTS

• **Human-centered computing** → **Mobile computing; Ubiquitous and mobile computing systems and tools**; Mobile devices;
• **Applied computing** → **Physics**; • **Computer systems organization** → **Embedded and cyber-physical systems**; • **Hardware** → **Sensor applications and deployments**.

KEYWORDS

Oral health; Toothbrushing monitoring; Dental plaque indication; Visible light channel modeling; Visible light sensing;

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1 INTRODUCTION

Motivation: Effective plaque removal is one of the core issues in oral hygiene, as it helps prevent dental caries, calculus, and gum disease [1, 2]. As the clinical gold standard, harmless plaque disclosing agents are widely used in dental practice [3]. These agents can stain plaque a purplish-red or purple, making it visible. More commonly, plaque disclosing agents are also recommended by over 50% dentists for use by the general public before brushing, particularly for individuals who need extra attention to their oral hygiene, e.g., those with severe dental issues like caries or periodontitis, children and the elderly with limited brushing skills, or patients with braces, to help them thoroughly remove plaque [4, 5]. Importantly, beyond these specific populations, these agents offer population-wide preventive benefits by enhancing plaque control efficacy and consequently reducing dental disease incidence.

However, using a plaque disclosing agent before brushing presents challenges. After using the disclosing agent, while plaque on the outer surfaces of the front teeth and the occlusal surfaces of molars can be seen with the naked eye, it is difficult to see the inner surfaces of the front teeth and the inner and outer surfaces of the molars without the assistance of an oral endoscope or mouth opener [6, 7]. This not only requires multiple rinses to remove foam but also necessitates frequent switching between the toothbrush and the endoscope or mouth opener to ensure thorough plaque removal. This process is not only inconvenient but also significantly increases brushing time [6]. These practical usage gaps diminish the utility of plaque-disclosing agents and hinder efficient plaque removal.



Fig. 1. PIT Proposal: After using a plaque disclosing agent, PIT provides real-time plaque visualization during brushing even with foam obstruction, helping users locate and remove plaque more effectively.

Our Approach: In this paper, we present PIT, a novel rotating electric toothbrush designed to be used after applying a disclosing

agent, for real-time plaque indication during brushing. (Figure 1). PIT features a careful hardware design, including a PCB with LEDs and a micro camera, along with an innovative toothbrush structure. It combines plaque visibility improving technology with a deep learning (DL) model to penetrate toothpaste foam and reveal plaque patches, providing real-time plaque feedback during brushing, even when thick foam is present. The development of PIT helps fully utilize the role of plaque-disclosing agents to bridge the aforementioned gaps and achieve higher oral cleanliness through efficient and targeted brushing. Note that, the recommended PIT protocol involves applying plaque disclosing agents before brushing, which is the more common practice [8] and motivates targeted plaque removal [9]. In contrast, post-brushing application serves to evaluate brushing efficacy [8] and requires additional brushing to address residual plaque.

Challenges and Solutions: Designing PIT, however, presents two main challenges. The first challenge is related to the toothbrush hardware design. Unlike existing toothbrushes, we need to add a micro camera for capturing close-up images of the teeth and LEDs for illuminating the dim oral cavity. The challenge is how to deploy the new hardware and design the toothbrush structure to ensure the stability of the newly deployed LED light source and micro camera, while also providing the mechanical vibration of bristles in a comfortable and discreet form ready for production. To address this, we designed a circular PCB with a hollow shaft attached to its backside, and the center of the bristles is hollowed out to allow the circular PCB to pass through. The hollow shaft of the PCB is rigidly connected to the stable supporting shaft of the shell using a mortise and tenon structure, ensuring that the PCB remains relatively stable with respect to the toothbrush while the bristles rotate along the axis.

Secondly, during brushing, the foam generated by the toothpaste covers the surface of the teeth, masking the stained plaque patches underneath. To preserve the purpose of the plaque disclosing agent as a brushing indicator, PIT needs to have the ability to 'see through' the toothpaste foam. However, the varying amounts and uneven distribution of foam make it challenging to detect these patches, presenting an ill-posed problem [10]. For this, we started by modeling the light propagation channels as the theoretical foundation for design guidance. We amplified the characteristics of the light path reflected by the plaque through the proposed plaque visibility improving technique (selecting a specific-wavelength green light source and enhancing the LED brightness). We also developed a GAN-based model to tackle thick foam areas, ultimately achieving robust and accurate plaque segmentation.

Summary of Experiment Results: We deployed PIT and evaluated its performance using a denture model, demonstrating its excellent plaque recognition capability (75.22% IoU (Intersection over Union)), real-time performance (29 ms latency), and low energy consumption (1.8–2.3W power). PIT maintains robust performance under various conditions, including different plaque disclosing agents, toothpaste colors, foam levels, and lighting conditions. Case studies also show that users can achieve better plaque-cleaning results with PIT without compromising comfort, compared to advanced electric toothbrushes.

Contributions: The primary contributions of this work are outlined as follows: (i) Firstly, we designed a new visual toothbrush

hardware with the camera and bristles overlapping, helping users effortlessly align the plaque. (ii) Secondly, we model the light propagation channel with the effect of foam and reflective surfaces (the teeth and plaque regions in this application). We believe this can provide general theoretical guidance and inspiration for more optical sensing applications. (iii) We designed an algorithm to identify plaque under foam, incorporating plaque visibility improving techniques and a lightweight deep learning network, enabling plaque visualization during brushing. (iv) Finally, we deployed PIT and evaluated its performance on dentures and human teeth, demonstrating its accurate plaque segmentation capability, real-time performance, robustness under various conditions, and higher plaque cleaning rate within the same time.

2 SYSTEM OVERVIEW

2.1 Design Goals

To advance the practical use of PIT, the system design must meet the following basic requirements:

- (1) **Comfort.** Integrating new electronic components into a standard electric toothbrush to enable plaque detection will alter its shape. To ensure broad user adoption, the design must maintain a similar comfort to the regular toothbrush. For example, it is essential to ensure that the mechanical structure does not cause friction against the oral cavity.
- (2) **Structural and Electrical Stability.** We need to ensure the new hardware components (i.e., micro camera and LEDs) are not affected by the vibrations of the toothbrush to guarantee the usability of image processing. Moreover, maintaining stable wire connections is crucial for long-term use.
- (3) **Robustness.** Since users will employ various brands of plaque-disclosing tablets or solutions, as well as different brands of toothpaste with varying colors and foam amounts, the system must be able to handle these diverse interference factors effectively to ensure usability.
- (4) **Accurate and Real-time Performance.** Leveraging the ubiquitous and readily accessible processing power of smartphones to handle images captured by the toothbrush camera can reduce costs and conserve computing resources. Providing highly accurate and fast plaque feedback on smartphones is essential for a good user experience.

2.2 System Overview

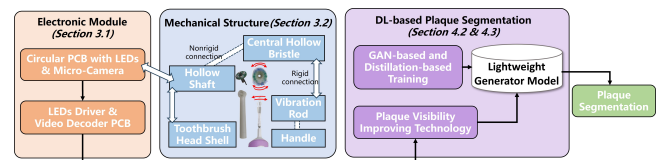


Fig. 2. The Overview of PIT.

Figure 2 shows the overall architecture of PIT, which consists of 3 main modules that bring together a practical system.

(1) **Visual Perception Toothbrush Electronic Module.** The designed additional electronic hardware mainly consists of two PCBs: the circular PCB, equipped with a micro camera and LEDs, is used to illuminate the oral cavity and capture images, while the LEDs

driver and video transmission PCB is responsible for driving the LEDs and transmitting the captured images to a smartphone.

(2) Mechanical Structure. New designed toothbrush mechanical structure works with the electronic hardware to create a toothbrush with rotating bristles but stable camera view.

(3) DL-based Plaque Segmentation Module. By integrating the LEDs light source design and processing the captured video using a DL algorithm, PIT can accurately and robustly identify dental plaque.

3 HARDWARE DESIGN

This section aims to complete the system hardware design, both electronic and mechanical, based on the design requirements outlined in Section 2.1. The electronic design defines the hardware architecture based on system functions and technical requirements, while considering the convenience of the mechanical design. The mechanical design must address comfort, stability, aesthetics, and hardware layout. Throughout this process, five researchers held multiple discussions, and after several iterations, the system reached its final form.

3.1 Electronic Design

3.1.1 Electronic Component Selection. The selection and coordination of electronic components are crucial for system functions and user experience. To meet the requirements of effectiveness, reliability, and assembly, we have chosen two key electronic components. **Micro camera.** To capture visual information, a camera is essential. After careful consideration, we selected the medical endoscope lens OVM6946-RAJA [11]. Firstly, its compact size (1.1mm×1.1mm×2.3mm) fits the limited space of the toothbrush head. Secondly, its depth of field of focus ranges from 5 mm to 50 mm, covering the distance between the toothbrush and the teeth during brushing. Thirdly, its 120° wide-angle field of view (FOV), allows for a broader view of the teeth. Fourthly, its 160K-pixel (400×400) resolution and 30 fps ensure real-time high-definition image transmission. Additionally, it has low power consumption and is non-toxic.

LEDs. Due to the lack of light in the oral cavity, providing a light source is essential. We chose the narrowband green LED with a wavelength range of 510-530 nm and analyzed the selection reasons in Section 4.2.1. The LED has a 120° uniform light emission angle. We chose the smallest package type, 0402 (1.0mm×0.5mm×0.4mm), for easy integration into the micro PCB on the toothbrush head.

3.1.2 PCB Design.

A circular PCB deployed with micro camera and LEDs. To fit the shape and size of the round toothbrush head, the circular PCB's diameter was only 8mm, leaving most of the bristle area unobstructed and facilitating subsequent designs that allow it to insulate the bristle rotation. A 1.2mm×1.2mm square hole in the center holds the micro camera, with four symmetrically placed LEDs around it. Note that the camera is slightly higher than the LEDs by 2mm to prevent direct light interference. The PCB is a double-layer design, using through-hole technology for wiring the LEDs and camera from the back, facilitating subsequent wiring. We wrapped the micro camera and LED wires together, encasing them in an insulating layer with a diameter of approximately 0.8mm. Notably, the PCB surface and junctions are sealed with food-grade

transparent epoxy resin for waterproofing and a smooth hydrophobic surface (Figure 3(a)). During manufacturing, a smooth plastic casing can be used to encase the circular PCB.

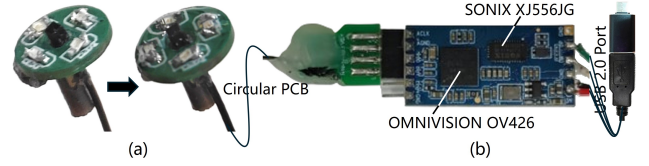


Fig. 3. (a) Epoxy resin sealing treatment of circular PCB. (b) LEDs driver and video transmission PCB.

LEDs driver and video transmission PCB. As shown in Figure 3(b), we designed a separate PCB connected to the aforementioned circular PCB to power the LEDs and transmit the camera video via a USB-compliant interface. Specifically, we used an OV426 camera bridge processor [12], which features a built-in ADC to digitize the analog output from the micro camera. The data was then processed by a digital signal processor (DSP). Next, we used a XJ556JG controller [13] to transmit the data at a rate of 30fps to the USB 2.0 interface. The LEDs were powered via the USB interface's 5V power supply and a 28 Ω resistor.

Note that, PIT's hardware cost is around \$41, comparable to \$35 cost of iPMTC toothbrush's visualization module [14].

3.2 Mechanical Design

3.2.1 Structure Composition and Manufacture. Note that while placing the camera a certain distance below the brush head avoids the bristles' rotation, the FOV is misaligned with the bristles, making it harder for users to adjust the brushing area based on proprioception, as users typically rely more on visual cues [15–17]. Plus, it may fail to capture deeper areas of the mouth.

If the circular PCB is simply placed at the center of the bristles, it will rotate with the bristles, causing the captured image to become significantly blurred and making it difficult to identify plaque. Moreover, we need to ensure that the wire are not exposed and maintain its stability to enhance durability. Furthermore, we must avoid introducing sharp mechanical structures that may rub against the user's mouth to ensure comfort.

To address the challenges and meet the design requirements, we developed the following mechanical structure, which includes four main parts as shown in Figure 4: (1) a circular PCB with a hollow shaft, (2) a central hollow bristle section, (3) a shell with dual cavities, and (4) a toothbrush handle with a horizontally linear vibrator. Specifically, we disassembled an Oral-B ProWhite toothbrush [18] and reused the bristles and the handle with vibrator. Using a plastic heat staking tool, we created a 3mm hole in the center of the bristle section. The umbrella-shaped hollow shaft and cavity shell were modeled in 3D Studio Max and 3D-printed with acrylonitrile-butadiene-styrene material via stereo lithography appearance technology, providing excellent rigidity and waterproofing.

3.2.2 Structural Connection and Assembly. Figure 5 shows the entire assembly process. As shown in Figure 5(a), the cavity shell is fitted in a stable position of the handle below the vibrator, and the diameter of the large cavity inside the shell is greater than the vibration range of the vibrator, so the brush head cavity is unaffected

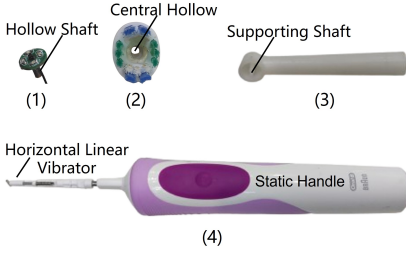


Fig. 4. PIT's structure components: (1) a circular PCB with a hollow shaft attached to its backside, (2) a central hollow bristle section, (3) a shell with a supporting shaft and dual cavities structure, and (4) a toothbrush handle with a horizontal linear vibrator.

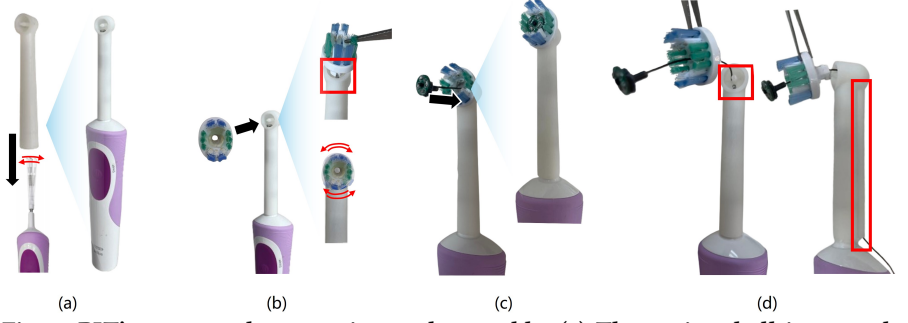


Fig. 5. PIT's structural connection and assembly. (a) The cavity shell is securely positioned below the vibrator on the handle. (b) The bristle section is mounted on the toothbrush head's shaft, aligning with the handle's vibrator. (c) The hollow shaft of the circular PCB is then mounted on the shell's supporting shaft. (d) PCB wiring is routed close to the rotating shaft, away from the vibrator, through a dedicated conduit to ensure electrical stability.

by the vibrator's vibrations. As shown in Figure 5(b), the bristle section is mounted on the supporting shaft of the toothbrush head cavity shell, and the groove of bristle section is fitted onto the vibrator of the toothbrush handle. Utilizing the central shaft limiting structure of the cavity shell, the horizontal linear vibration of the handle's vibrator is converted into rotational motion of the bristle section. The circular PCB was glued to the umbrella-shaped hollow shaft structure and formed an umbrella-shaped circular PCB section. Then, this umbrella-shaped circular PCB section is mounted onto the shell's supporting shaft using a tenon and mortise structure, forming a rigid connection, thus keeping the umbrella-shaped structure with the camera and LEDs relatively stable in relation to the brush handle (Figure 5(c)). The camera is in the center of the bristles and they are very close together, facilitating the user to place the bristles on the plaque quickly and naturally after seeing the plaque in the camera's field of view based on visual information and proprioception [15–17]. The micro camera is approximately 5mm from the bristle tips. During brushing, it captures an image about 8mm wide, providing a sufficiently large FOV to see 1 to 2 teeth.

In addition to meeting the requirement that mechanical vibrations do not affect image blurriness, we also need to ensure electrical stability. Specifically, we need to keep the fine wires connecting the circular PCB section and the LEDs driver and video transmission PCB free from interference by the vibrator. To this end, the cavity shell features a large conduit that accommodates the vibrator and a separate narrow conduit for the wires. As shown in Figure 5(d), the wire routing is close to the rotating shaft, away from the vibrator below, and passes through the dedicated conduit. This way, vibrations from the vibrator in the large conduit do not impact the wires in the narrow conduit.

4 PLAQUE SEGMENTATION ALGORITHM

During brushing, toothpaste foam gradually covers the teeth, making it challenging to sense plaque beneath. This is a new issue introduced by the application, which has not yet been studied. To address this, we modeled the effect of foam and the tooth surface

on light, which led us to propose corresponding technical solutions and ultimately achieve robust plaque recognition.

4.1 Modeling Light Channels with the Influence of Foam and Teeth (Plaque)

In this section, we analyze the light channel and model the captured images to lay the theoretical foundation for the PIT algorithm.

4.1.1 Foam Structure. We used a toothbrush, foamy toothpaste, and a small amount of water to rub on a black acrylic plate for about ten seconds to generate foam, and then observed the foam under a microscope at different magnifications. At the micro level, foam consists of liquid membranes enclosing air, stacked together (Figure 6(a)), with liquid channels (edges) filling the space between adjacent bubbles (Figure 6(b)). In 2D, the foam includes **ultra-thin films** and **thin edges**.

4.1.2 Single Light Channel Model. Due to the proximity of the four LEDs, the light from them combines and is perceived as a virtual point light source located at the center (green point in Figure 7(b)). The camera is also treated as a red point in Figure 7(b) coinciding with the virtual point light source for easy analysis. Due to the lens's focusing effect, the CMOS sensor's pixels (corresponding to image pixels) only receive light from specific directions, allowing us to simplify the analysis by considering the light received by a single pixel. Assume a CMOS sensor's pixel receives light from the black dashed direction in Figure 7(b). The point light source emits a beam in this direction, and we analyze all light paths along it. Note that light from other directions is ignored as it cannot be focused on this pixel.

As shown in Figure 8(a), assume that the ultra-thin film contains a very small number of dye molecules, so it has a negligible effect on the light. The edge transmits, reflects, and absorbs fixed proportions of the light. When light passes through the edge, the refracted angles cancel due to opposite media at the two interfaces. Since the edge is still thin, the light after penetration remains in the original direction (i.e., two black dashed lines as shown in Figure 8(b) coincide). Light attenuation in air is neglected, as it is much weaker than the edge's blocking attenuation.

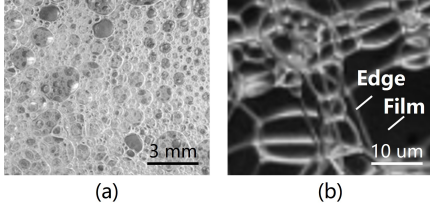


Fig. 6. Microscopic views of the toothpaste foam structures at (a) millimeter and (b) micrometer scales under optical microscopy.

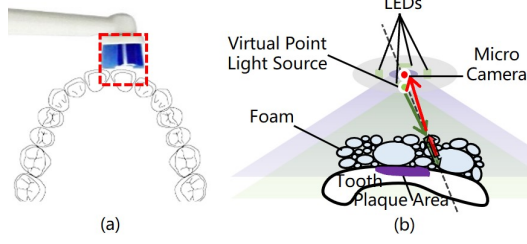


Fig. 7. (b) We analyze the light path of a single ray along the dashed line (a) after the toothbrush contacts the teeth.

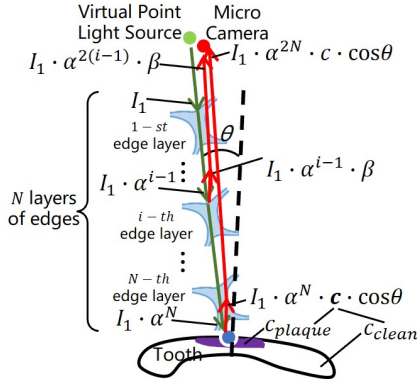


Fig. 9. Single light propagation model.

4.1.3 Light Intensity Function of a Single Pixel. Assume the light beam to be analyzed penetrates N layers of edges of foam on the tooth surface (Figure 9). The light intensity from the point source is I_1 , which is also when it reaches the first edge layer, as we ignore the air attenuation. Then, the light intensity I_i when the light reaches the i -th layer of edge after transmitting $i-1$ layers can be expressed as $I_1 \cdot \alpha^{i-1}$, where α is the transmission coefficient of the edge. This i -th layer reflects $I_1 \cdot \alpha^{i-1} \cdot \beta$, where β is the reflection coefficient, and the intensity of the reflected light returning to the first layer (i.e., CMOS sensor pixel) after passing through the $i-1$ layers of edges is then $I_1 \cdot \alpha^{2(i-1)} \cdot \beta$. Thus, The total reflection effect of the N layers of edges can be expressed as:

$$R_{edge,N} = \sum_{i=1}^N I_1 \cdot \alpha^{2(i-1)} \cdot \beta = I_1 \cdot \frac{1 - \alpha^{2N}}{1 - \alpha^2} \cdot \beta \quad (1)$$

The light that passes through the N -th layer of edge is reflected by the tooth surface. Because the tooth reflection can be considered to conform to Lamber's law [2], that is, the reflected light intensity is proportional to the cosine of the angle between the surface normal and the incident light [19], the intensity of the light reflected back to the first edge layer is:

$$R_{tooth,N} = I_1 \cdot \alpha^{2N} \cdot c \cdot \cos \theta \quad (2)$$

where c is the tooth reflectivity, which is ≥ 0 and ≤ 1 , varying between stained plaque and clean surfaces. The reflectivity of the stained plaque patches is denoted as c_{plaque} and the reflectivity of the clean tooth area is denoted as c_{clean} . θ is the angle between the normal of reflection point (blue point in Figure 9) on the tooth

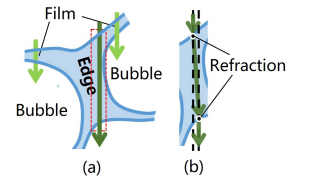


Fig. 8. (a) Only edge transmission is analyzed and (b) transmitted light is assumed to remain in the original direction.

surface and the incident light, which is less than 90° . Therefore, the total light intensity reaching the camera pixel is:

$$R_{total,N} = R_{edge,N} + R_{tooth,N} = I_1 \cdot \frac{1 - \alpha^{2N}}{1 - \alpha^2} \cdot \beta + I_1 \cdot \alpha^{2N} \cdot c \cdot \cos \theta \quad (3)$$

In this function, α and β are constants where $0 < \beta < \alpha < 1$ and $\alpha + \beta < 1$. α and β change only when switching between different toothpastes, due to the concentration of dye molecules in the toothpaste. With a fixed light source intensity I_1 , different regions in an image have different edge layers N , and different pixels correspond to different θ . The reflectivity c varies between plaque-covered areas c_{plaque} and clean tooth surfaces c_{clean} . Note that, mixing foam with blood/solution turns it light red/purple, reducing α . This mimics increasing N , which is subsequently analyzed and doesn't change conclusions.

4.1.4 Numerical Simulation and Analysis. Next, we analyze how the variables affect foam edge reflection and tooth reflection changes and total received light intensity in Equation (3). The edge transmission coefficient α is set to 0.8, the reflection coefficient β to 0.1, the plaque coverage reflection coefficient c_{plaque} to 0.1 or 0.5, the clean tooth surface reflection c_{clean} to 1, and the θ is set to 0 when analyzing other variables. Note that the exact values do not affect the analysis conclusions since we are focused on trends.

Analysis 1: Effect of Edge Layers N on Foam Reflection

$R_{edge,N}$ and Tooth Reflection $R_{tooth,N}$: As shown in Figure 10(a), with N increases, the foam effect part (green curve) increases from 0 to nearly $I_1 \cdot \frac{1}{1 - \alpha^2} \cdot \beta$, while the tooth reflection decreases from $I_1 \cdot c \cdot \cos \theta$ to nearly 0. As shown in Figure 10(a), when N is smaller than the red intersection point of $R_{tooth,N}$ curve and $R_{edge,N}$ curve, the tooth reflection $R_{tooth,N}$ is greater than the foam reflection $R_{edge,N}$. This suggests that in areas with less or sparser foam, the teeth have a greater impact on the CMOS sensor pixels, and a higher signal-to-noise ratio (SNR) facilitates recognition.

Analysis 2: Effect of Plaque Reflectivity c_{plaque} on Total Reflection $R_{total,N}$

$R_{total,N}$: As shown in Figure 10(b), the reflectivity c_{plaque} in the plaque patches is lower than the reflectivity c_{plaque} in the clean area, resulting in a difference in total light intensity $R_{total,N}$. However, as can be observed from Figure 10(b), when the plaque reflection coefficient c_{plaque} is 0.5, the curves of $R_{total,N}(c_{plaque} = 0.5)$ and $R_{total,N}(c_{clean} = 1)$ both show a decreasing trend. At the position of the red dashed line, the total received light intensity $R_{total,N}$ in both plaque and clean areas is the same for different foam thicknesses (edges) N , reducing the robustness of intensity-based recognition. However, when c_{plaque} is

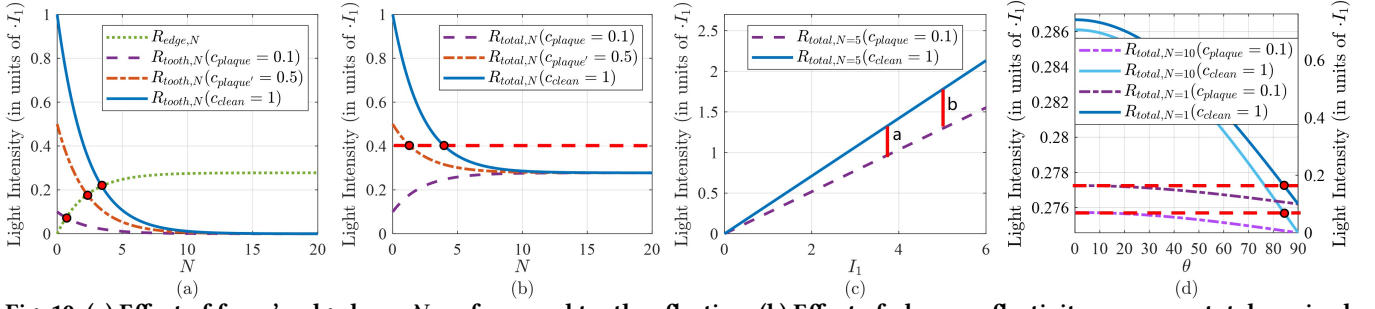


Fig. 10. (a) Effect of foam's edge layer N on foam and tooth reflection. (b) Effect of plaque reflectivity c_{plaque} on total received light intensity $R_{total,N}$. (c) Effect of light source intensity I_1 on $R_{total,N}$. (d) Effect of angle θ on $R_{total,N}$.

smaller (e.g., 0.1), the total intensity $R_{total,N}$ shows distinct trends in these two areas, aiding robust recognition. This inspires us to find ways to reduce the plaque patches' reflection coefficient c_{plaque} .

Analysis 3: Effect of Light Source Intensity I_1 : When the light source intensity I_1 increases, the total received light intensity in both the plaque and clean tooth areas also increases. However, due to the different rates of increase, a greater difference arises between them, which aids in recognition (The length of line a in Figure 10(c) is greater than that of line b). This encourages us to increase the LED light intensity.

Analysis 4: Effect of Angle Between Tooth Reflection Point Normals and Incident Light θ : As shown in Figure 10(d), when the angle between the tooth surface normal and the direction of the incident light θ is large, such as when encountering interdental spaces, the total reflected light intensity from the clean areas of the tooth decreases. However, we observe that when θ is less than approximately 85° , the total reflected light intensity from the clean tooth area is consistently greater than that from the plaque patches, which is sufficient for differentiation. More importantly, we find that further reducing the plaque reflectivity c_{plaque} can expand the distinguishable θ range. This suggests that the effect of θ can be negligible, and motivates us to reduce the plaque reflectivity c_{plaque} .

4.2 Model Validation and Light Source Design for Improving Plaque Visibility

The previous analysis suggests reducing c_{plaque} and increasing light intensity I_1 to make the plaque patches' intensity lower than the clean area and maximize the difference for better recognition.

4.2.1 LEDs Wavelength Selection. The white clean tooth surface reflects all wavelengths, while the reflection coefficient of dental plaque c_{plaque} varies with the wavelength of the light source. According to previous studies [20, 21], colored objects tend to almost completely reflect the light of the same color, while selectively absorbing certain wavelengths and reflecting little or no light. Thus, we compared six different LED light sources (white, yellow, green, orange, red and blue; Figure 11) and we found that the green light source provided the best contrast. This is because the purplish-red dental plaque reflects the least green light (c_{plaque} is minimized), making the plaque patches appear almost black, enhancing the distinction between the clean tooth and plaque patches. Additionally, the CMOS sensor's high sensitivity to green light results in high

image clarity [22]. Through this light source wavelength selection, we minimized the dental plaque reflectivity c_{plaque} .

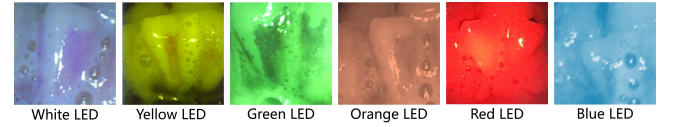


Fig. 11. Images captured with LEDs of different colors.

4.2.2 LEDs Intensity Enhancement. Building on the previous analysis, we know that increasing light intensity enhances the contrast between the reflected light intensities of the clean area and plaque patches, which improves dental plaque detection. However, we must also consider the light intensity range that the camera's CMOS sensor can handle. This is to prevent overexposure, which can cause the loss of brightness details in the image. After selecting the green LED, we applied varying brushing forces and adjusted the light intensity. Figures 12(b)(c)(d) show the corresponding results when the light source drive voltage was increased from 2.3V to 2.76V, and then to 3.0V (overexposure).

To validate the correctness of the model in Section 4.1 and the effectiveness of light source selection in the previous section, we selected seven sampling points with fewer edge layers from both the plaque and clean areas, as shown in Figures 12(b)(c). Figure 13 displays the G channel values from the RGB images of the sampling points. The results show that, when light intensity is fixed, the G values in the plaque patches are almost always smaller than those in the clean area. This supports the effectiveness of using a green light source to minimize c_{plaque} , which is also consistent with Analysis 2 in Section 4.1.4. As the light source voltage increased from 2.30V to 2.76V, the light intensity in the clean area (left red box in Figure 13) increased more than in the plaque patches (right red box in Figure 13). This resulted in a larger intensity difference between the clean and plaque areas, which is also in line with the conclusions of Analysis 3 from Section 4.1.4 regarding the impact of light source intensity. Therefore, we used a $28\ \Omega$ resistor to divide the 5V power supply, setting the voltage across the LEDs to 2.76V, achieving the maximum brightness before overexposure.

Through this section, we optimized the light source intensity to maximize the contrast between the plaque and clean areas in regions with less foam. At the same time, we validated the correctness of the theoretical modeling and simulation analysis through practical experiments.

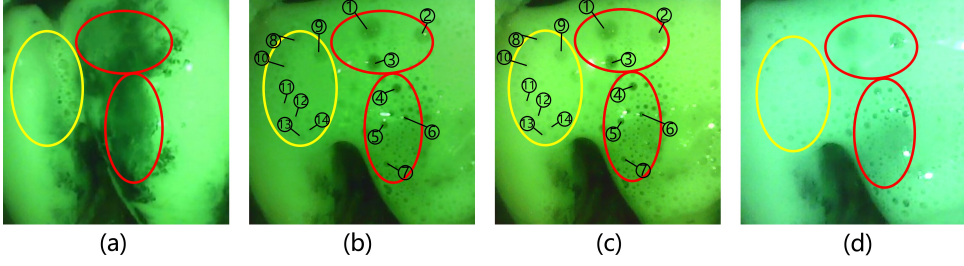


Fig. 12. (a) Ground truth image without foam coverage (the red area represents dental plaque, and the yellow area represents clean teeth). (b) Image with 2.30V LEDs drive voltage. (c) Image with 2.76V LEDs drive voltage. (d) Overexposed image.

4.3 DL-based Plaque Segmentation

The previous section improved differentiation of plaque and clean teeth under sparse foam area. However, as Analysis 1 in Section 4.1.4 shows, perceiving the tooth surface in dense foam remains challenging. Inspired by Context Encoders [23], we introduce the idea of filling in missing pixels in the image, where areas occluded by thick foam can be considered as "missing" pixels, whereas regions with sparse foam are considered as present. Thus, we propose a deep learning model to combat this challenge and generate plaque region segmentation masks.

GAN-based Network Design. Our goal is to train a generator to create an accurate plaque segmentation binary mask. We chose the U-Net architecture [24] as the foundation for the generator because it is well-suited for image-to-image generation tasks. More importantly, its symmetric encoder-decoder structure, enhanced by skip connections, allows the transmission of both low-level spatial details and high-level semantic information, which is crucial for accurate plaque segmentation. To improve the model's training efficiency and generalization capability, we employed a zero-sum game strategy to guide the generator. For the discriminator design, we drew inspiration from PatchGAN [25]. Instead of classifying the entire image as true or false, PatchGAN operates on an $k \times k$ (set to 48) "receptive field" within the generated image and penalizes it. Specifically, the discriminator judges the authenticity of each $k \times k$ patch in the image, convolving this discriminator across the image and averaging the results as the output. The benefit of this approach is that it guides the generator to focus on local plaque segmentation.

Online GAN Distillation. Since we expect users to run our model inference on a ubiquitous smartphone to watch plaque segmentation in real-time, we need to ensure the trained generator model is lightweight. We adopt the knowledge distillation method to obtain a compressed student model. Inspired by the training techniques of OMGD described in [26], we decouple the student generator from the discriminator, allowing it to directly learn from the teacher model, as this facilitates flexible training and significant compression. The architecture and parameters of the entire training network are shown in Figure 14. In this setup, we create a deeper and wider teacher model and a student model with fewer parameters. To ensure that the student model G_S closely mimics the teacher model G_T and generates high-quality segmentation, we integrate multiple loss components, including structural similarity loss $\mathcal{L}_{SSIM}(G_T, G_S)$ [27], feature loss $\mathcal{L}_{feature}(G_T, G_S)$ [28], style loss $\mathcal{L}_{style}(G_T, G_S)$

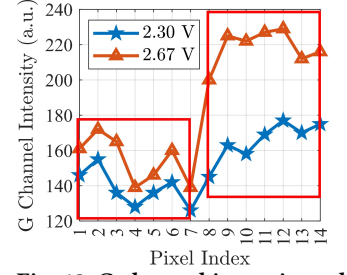


Fig. 13. G channel intensity values at different sampling points in Fig. 12(b) and (c).

[29], and total variation loss $\mathcal{L}_{TV}(G_T, G_S)$ [30], to quantify the discrepancy between the student and teacher models. Therefore, the knowledge distillation loss (KD loss) is defined as:

$$\mathcal{L}_{KD}(G_T, G_S) = \lambda_{SSIM} \mathcal{L}_{SSIM}(G_T, G_S) + \lambda_{feature} \mathcal{L}_{feature}(G_T, G_S) + \lambda_{style} \mathcal{L}_{style}(G_T, G_S) + \lambda_{TV} \mathcal{L}_{TV}(G_T, G_S) \quad (4)$$

where λ_{SSIM} , $\lambda_{feature}$, λ_{style} , and λ_{TV} are four hyperparameters used for balancing.

GAN Loss Function Design. In the minimax game between the teacher model and the discriminator, We designed a combined loss function (adversarial losses and reconstruction losses) to jointly optimize the generator and the discriminator. The adversarial loss encourages the generator to produce masks that the discriminator cannot distinguish from ground truth. We adopt a conditional GAN objective:

$$\mathcal{L}_{GAN}(G_T, D) = \mathbb{E}_{\mathbf{x}, \mathbf{y}} [\log D(\mathbf{x}, \mathbf{y})] + \mathbb{E}_{\mathbf{x}} [\log (1 - D(\mathbf{x}, G_T(\mathbf{x})))] \quad (5)$$

where $\mathbf{x}$ is foam-covered input and $\mathbf{y}$ is groundtruth plaque mask. Objective of teacher generator G_T is to minimize this objective, while training objective of discriminator D is to maximize it.

To ensure precise pixel-level predictions, we incorporate a pixel-level L_1 reconstruction loss:

$$\mathcal{L}_{L1}(G_T) = \mathbb{E}_{\mathbf{x}, \mathbf{y}} \|\mathbf{y} - G_T(\mathbf{x})\|_1 \quad (6)$$

This term penalizes the generator whenever it deviates from the ground truth segmentation. Combining above these two objectives, the final objective is:

$$G^* = \arg \min_G \max_D \lambda_{GAN} \mathcal{L}_{GAN}(G_T, D) + \lambda_{L1} \mathcal{L}_{L1}(G_T) \quad (7)$$

where λ_{GAN} and λ_{L1} control the relative contribution of each loss term.

5 SYSTEM EVALUATION

In this section, we present the experimental setup and a series of evaluations conducted to assess the system's performance. We begin by evaluating the overall performance of the PIT dental plaque recognition algorithm, followed by ablation studies to assess the effectiveness of different technology components. Finally, we examine other factors that may affect PIT's performance, highlighting its robustness.

5.1 Experimental Setup

Dataset Collection. To evaluate plaque detection accuracy under foam, we needed both PIT-captured images during brushing and

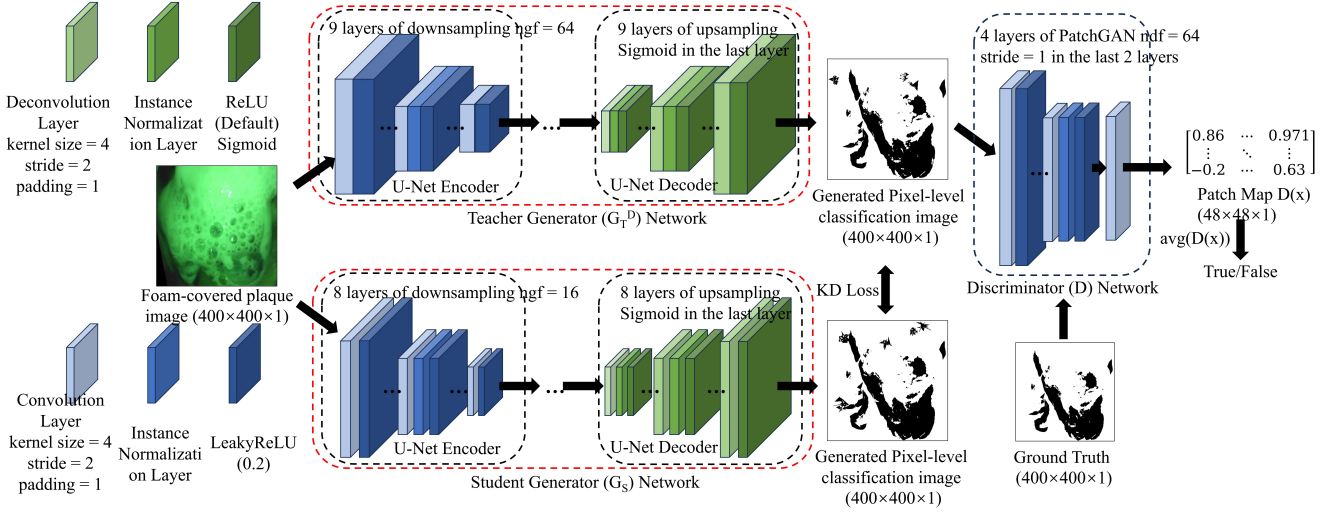


Fig. 14. GAN-based and distillation-based model overview.



Fig. 15. Denture model and close-up of PIT.

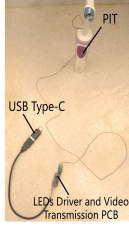


Fig. 16. Segmentation results.

		73.25 %	93.78%	85.97%
Captured Image				
Recognition Result				
Ground Truth				

corresponding ground truth images. However, the foam from toothpaste makes it difficult to obtain ground truth visually. Thus, we attempted to first collect ground truth images, and then perform brushing to collect foam images. However, we found that plaque removal during brushing prevents accurate alignment with the initial ground truth. To overcome this issue, we chose to use a denture model and applied seven shades of purplish-red oil-based paint to simulate the stained plaque (Figure 15(left)). We then applied polymeric nanocoating to the denture surface, forming a waterproof film. Note that although the coating provides good wear resistance and protective properties, it cannot be directly used on human teeth due to the contained toxic fluoride. We first collected ground truth images by turning on only the green LEDs. Then, using PIT with water and a custom medium-foam toothpaste (1.5 wt% sodium lauryl sulfate (SLS)), we simulated the brushing process with activated LEDs and bristle rotation. We manually selected four pairs of corresponding feature points and aligned the images using homography matrix transformations to ensure alignment quality. Meanwhile, a manual color thresholding technique was applied to the ground truth images to segment the plaque region. We changed the distribution of the "plaque" 8 times and obtained a total of 400 labeled images. We separately changed the LEDs to white and yellow and set the driving voltage to 2.30V, then collected data using the same procedure for the ablation experiment for subsequent studies.

Detailed Training Settings for PIT model. The teacher model G_T was initialized and trained with a batch size of 32 and an initial learning rate of 10^{-4} . Cosine annealing was applied to progressively decay the learning rate to 10^{-6} over 20000 iterations. The adversarial loss weight λ_{GAN} with the discriminator D was set to 0.95. The smaller student generator G_S was trained using KD loss. During training, G_S used a batch size of 64 and an initial learning rate of 5×10^{-4} . To stabilize the optimization process, cosine annealing reduced the learning rate to 10^{-6} over the first 1000 epochs. To avoid local optima caused by annealing schedules, the learning rate was reset to a constant 5×10^{-5} and training was continued for an additional 500 epochs. All models were implemented in PyTorch and trained on two NVIDIA RTX 4090 GPUs. Adam optimizers were used for G_T , G_S , and D with $\beta_1 = 0.5$ and $\beta_2 = 0.999$. After the training was completed, the student model was stripped down and used as the PIT model running on the mobile device. Note that, the model can run locally and offline, eliminating the need for cloud uploads and alleviating privacy concerns.

Metrics. We evaluate the performance of the teacher model and the PIT student model by calculating the average IoU (Intersection over Union) through cross-validation. We also assessed the running speed and energy consumption on the Xiaomi 14 Ultra smartphone.

Table 1: Performance of different models on Xiaomi 14 Ultra (Snapdragon 8 Gen 3).

Implementation	Size	Inference Time	IoU	Power
Teacher Model (3-thread CPU)	58.59 M	101.92 ms	79.17%	3.1–3.7 W
Student Model (3-thread CPU)	2.88 M	37.83 ms	75.25%	2.3–2.8 W
Teacher Model (GPU)	58.59 M	35.74 ms	79.36%	2.0–2.4 W
Student Model (GPU)	2.88 M	12.64 ms	75.22%	1.8–2.3 W
LumiO [6] (4-thread CPU)	0.46 M	15.87 ms	34.61%	2.1–2.6 W

5.2 Plaque Identification Performance

Baseline accuracy and Intuitive visual results. The baseline accuracy of the PIT model is evaluated using 5-fold cross-validation. As shown in Table 1, the teacher model achieved an average IoU of 79.36% (GPU), indicating good performance when trained and tested on different and balanced subsets of the dataset. The compressed student model also achieved a comparable IoU of 75.22% (GPU) to the original model. Figure 16 presents several plaque recognition examples from the student lightweight model, visually demonstrating the similarity between PIT’s predictions and the actual plaque conditions. This suggests that PIT can accurately estimate the plaque condition beneath the foam despite the toothpaste foam interference. In contrast, LumiO’s KNN ($k=11$) achieved only 34.61% IoU due to its inability to reliably infer plaque under foam coverage.

Recognition Latency. As shown in Table 1, the student model running on Xiaomi 14 Ultra achieves inference latencies of 37.83 ms (CPU, 3 threads) and 12.64 ms (GPU). We adopted the student model (GPU) as the default configuration for PIT. When image transmission delays are included, the measured end-to-end latency averages approximately 29 ms, which is below the threshold of human perception and fully satisfies PIT’s real-time visual interaction requirements.

Power Consumption. We used the smartphone’s built-in energy consumption estimation tool to measure the energy consumption. The calculated power is shown in Table 1. Due to its reduced computational load, the student model results in lower system energy consumption, at only 2.6W (CPU) and 2.0W (GPU) on average, comparable to the 2.5W power consumption when watching full-screen 480P videos on the Bilibili APP. To examine the energy consumption distribution, we also measured the PIT hardware’s energy consumption using a current meter, which showed a current of 174mA. Therefore, with a 5V voltage, the power consumption is 0.87W. This indicates that the power consumption of driving LEDs and video transmission part is minimal, and the primary energy consumption comes from the DL model inference.

5.3 Ablation Experiment

This section investigates the effectiveness of two plaque visibility improving techniques in light source designs proposed in Section 4.2 through ablation experiments.

Effectiveness of LEDs Wavelength Selection. As shown in Figure 11 and Section 4.2.1, using green light provides the best contrast because the purplish-red plaque patches absorb green light almost completely, appearing black visually. In contrast, orange, red, and blue LEDs offer poor contrast. We calculated the accuracy of the data collected using white and yellow light sources, which also provide relatively good contrast. Note that we changed the model’s input from the G channel data to RGB channel data for better color representation. As shown in Figure 17, the accuracy for white and yellow LEDs was 60.27% and 65.15%, respectively. This indicates that choosing green LEDs can improve recognition accuracy.

Effectiveness of LEDs Intensity Enhancement. Section 4.2.2 proposes the design of increasing light intensity, it is validated that it can provide greater contrast between plaque and clean teeth, which is beneficial for recognition (Figure 13). Here, we further

validated the improvement in accuracy brought by this technique. We reduced the LEDs driving voltage to 2.30V, which provided adequate brightness while ensuring image clarity. The experimental results in Figure 17 show that lower light intensity led to a lower IoU of 64.83%. This may be because greater contrast in areas with thinner foam helps the deep learning network better infer the plaque situation beneath thicker foam.

5.4 Robustness

In this section, we evaluate the robustness of our PIT system under various conditions, including different plaque disclosing agents, foam color and amount, and varying ambient light intensity. For each case, we collected 50 photos of different areas on the denture model. We trained the PIT model using all the data collected under default conditions and tested it on the newly collected data.

Impact of Plaque Disclosing Agent Color. We purchased five different brands of dental plaque disclosing agents: Ci Medical (solution), SiRuiNi (solution), Yandy (solution), Dearlyfly (solution), and Y-kelin (tablet). Through testing, we found that their colors were purplish-red, pink, purplish-red, red, and purple, respectively. To simulate these colors, we used oil-based paint with similar shades and used standard toothpaste to brush the denture model. Figure 18 shows the test results. Among them, the PIT for detecting the red plaque (representing Dearlyfly) had the lowest IoU of 67.71%, while the recognition accuracy for the other plaque colors was above 70%. These results suggest that, although the color differences in plaque disclosing agents may affect image contrast due to varying absorption of green light, the training set with up to seven plaque colors in Section 5.1 is sufficient to cover these variations, meaning the effect of the disclosing agents is expected to be minimal.

Impact of Toothpaste Color and Foam Amount. Toothpaste contains a surfactant, typically sodium lauryl sulfate (SLS), with concentrations usually ranging from 0.5 wt% to 2.5 wt% [31], which helps to remove dental plaque. We customized toothpaste with different colors and SLS concentrations for this study. Figure 19 explores the impact of different foam colors and SLS concentrations on the plaque detection performance. The color of the toothpaste does not appear to have a significant impact, likely because the pigment concentration after foaming is minimal, while low-foam toothpaste helps achieve a higher IoU compared to high-foam toothpaste. This is because, after the toothpaste foams, light can still pass through the less dense foam areas. Additionally, the significant contrast between dental plaque and clean teeth under green light allows the PIT to use the deep learning model to distinguish plaque patches effectively. The findings suggest that the overall detection remains reliable across different conditions of foam color and concentration.

Impact of Ambient Light. We evaluated the performance of PIT under four different ambient light conditions. The light intensity levels tested were: (1) 5 lux (dark bathroom), (2) 210 lux (bathroom in the morning without ceiling light), (3) 428 lux (balcony in natural light), and (4) 743 lux (bathroom with ceiling light at night). As shown in Figure 20, the experimental results indicate that PIT achieves the highest IoU at 76.12% in the darkest environment (5 lux), with performance gradually decreasing as light intensity increases. As shown in Figure 20, there is a slight decrease in IoU

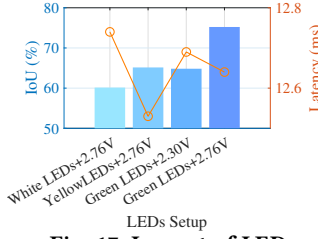


Fig. 17. Impact of LED wavelength and intensity.

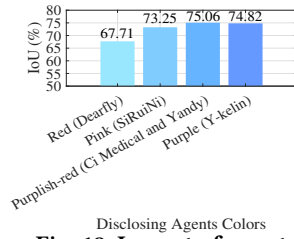


Fig. 18. Impact of agents brands' colors.

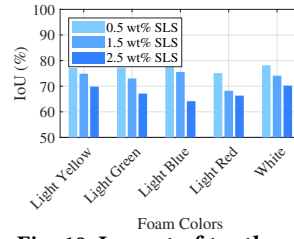


Fig. 19. Impact of toothpaste color and foam amount.

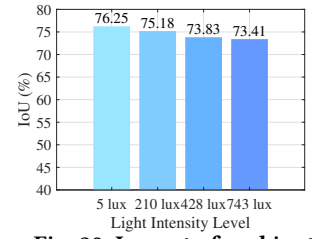


Fig. 20. Impact of ambient lighting condition.

(approximately 73%) under higher light intensities (428 and 743 lux). However, overall, PIT demonstrated adaptability to different environmental lighting conditions. The ambient light may illuminate the outer surface of the front teeth, while the light reaching other teeth is very weak [2]. Plus, we believe that a high-intensity green light source helps alleviate interference from ambient light.

6 HUMAN SUBJECT STUDY

In this section, we further demonstrate how users can interact with PIT and investigate its impact on users. We have obtained approval from the Institutional Review Board (IRB) for the human subject evaluation of the PIT system. The participants' experimental reward was the PIT toothbrushes they used in the experiment.



Fig. 21. Users use the PIT system to brush their teeth and watch real-time plaque feedback on the smartphone APP.

6.1 Experimental Setup

Participants. We recruited 10 participants (4 women and 6 men, aged 20 to 36) who were all non-smokers. Eight participants had existing resin-based dental restorations but presented no caries or periodontal conditions requiring intervention. Regarding oral hygiene habits, nine participants reported twice-daily toothbrushing while one maintained only evening brushing.

Comparative Devices. The experiment devices were (1) Oral-B iO9 [32] with real-time 16-area brushing coverage display, (2) iPMTC [14] with a camera beside the bristles for visual feedback, and (3) PIT toothbrush (PIT model was trained on all data collected under the default setting). PIT connects to the phone via USB Type-C (Figure 15), with users viewing plaque segmentation results on a dedicated app (Figure 21). We individually manufactured a PIT for each participant to ensure hygiene, while replacing the bristles for the other two types of toothbrushes and sterilizing.

Procedure. In this comparative study, we employed a within-subjects design to eliminate inter-participant variability in brushing technique and plaque levels. Specifically, each participant used all

three toothbrush types across separate nights, performing only one brushing session per night with a single toothbrush type. Before each session, participants were first asked to use Ci Medical plaque disclosing agent to form plaque stains. Specifically, each participant dropped 3–5 drops of the disclosing agent into a 10mL cup of water, swished it in their mouth for 20 seconds, and then spat it out. Afterward, they rinsed with clean water to remove any excess agent. We used an intraoral scanner to model the participants' mouths, and applied a manual color thresholding technique to identify plaque regions. Plaque was quantified using planar plaque analysis [33], calculating the percentage of plaque (in pixels) relative to the total tooth area (in pixels). Brushing was performed for 2 minutes with a custom-formulated toothpaste (1.5 wt% SLS content) while running the companion application on Xiaomi 14 Ultra smartphones (Figure 21). After brushing, we applied the same plaque quantification method to measure the remaining plaque percentage and calculated the plaque decrease percentage. Participants were asked for their opinion on the usability of the PIT compared to the other two devices after using all three toothbrushes (Table 3).

6.2 Results and Analysis

Table 2 shows the plaque reduction before and after brushing with three different toothbrushes for all participants. Before brushing, the plaque levels in all areas were similar, with no significant differences, indicating that the participants had comparable daily plaque accumulation. After brushing with the Oral-B iO9, the average plaque level decreased to 12.4%, with an average reduction of 64.7%. This could be because the Oral-B iO9 can only track 16 broad tooth areas and does not provide detailed visual information about plaque attachment on the teeth. When using iPMTC, the average plaque level decreased to 13.2%, representing a 62.3% reduction. However, iPMTC's plaque cleaning effectiveness was not superior to that of the non-visualized toothbrush (Oral-B iO9), especially the occlusal surfaces of the molars. We found that the camera located beneath the bristles had difficulty capturing the occlusal surfaces of the molars at the back of the mouth. Additionally, the camera's field of view only captured foam on the surface of the teeth, making it hard to see the plaque underneath. Therefore, the user's brushing strategy was largely blind. Although users could continue to remove plaque by brushing multiple times after rinsing, this would significantly increase brushing time. In contrast, when using PIT, the average plaque level dropped to 5.6%, with an average reduction of 84.5%. This demonstrates a significant improvement in both user perception and plaque removal ability with the PIT, highlighting the effectiveness of our hardware design and its ability

Table 2: Plaque reduction comparison between PIT and other commercial toothbrushes (mean percentage plaque coverage).

Tooth Surface	Before Brushing (%)			After Brushing (%)			Decrease Percentage (%)		
	Oral-B	iPMTC	PIT	Oral-B	iPMTC	PIT	Oral-B	iPMTC	PIT
Outer upper teeth	34.2 (31.5–37.1)	35.3 (30.8–38.4)	34.8 (30.2–36.4)	11.7 (9.1–14.1)	10.1 (8.5–13.1)	5.1 (1.7–8.2)	65.8	71.4	85.3
Outer lower teeth	35.3 (28.5–40.7)	34.6 (30.0–38.3)	36.5 (32.0–38.5)	12.2 (10.5–17.7)	11.5 (8.8–13.9)	5.3 (1.8–9.9)	65.4	66.8	85.5
Inner upper teeth	36.8 (30.4–42.7)	34.8 (26.1–40.1)	37.5 (32.5–41.9)	14.5 (10.7–20.1)	14.8 (10.3–21.8)	6.2 (2.1–11.1)	60.6	57.5	83.5
Inner lower teeth	33.5 (28.5–37.4)	36.2 (29.3–41.8)	35.6 (31.0–37.4)	12.6 (9.2–16.3)	13.2 (9.8–16.5)	5.7 (1.4–8.0)	62.4	63.5	84.0
Occlusal upper teeth	34.5 (31.7–35.2)	33.8 (28.8–36.4)	35.8 (32.6–37.0)	11.4 (8.1–14.3)	15.3 (12.7–21.3)	5.6 (1.2–9.5)	67.0	54.7	84.4
Occlusal lower teeth	35.4 (29.2–38.1)	35.3 (31.5–40.3)	36.9 (32.0–39.1)	11.8 (8.3–15.1)	14.2 (10.5–20.3)	5.8 (1.7–7.8)	66.7	59.8	84.3

to identify plaque beneath the foam. Beyond the aforementioned population-average analysis, we identified inter-user variations in brushing technique that directly influenced plaque removal efficacy. Three participants applying sequential brushing patterns achieved higher plaque removal rates (iPMTC/Oral-B: >70%; PIT: >85%). Additionally, seven users demonstrated 10% longer outer-surface dwell time, yielding higher plaque removal rates versus inner surfaces. Table 3 shows that PIT was perceived as equally comfortable as commercial toothbrushes. Despite reduced convenience scores (3.4 ± 0.52) due to agent use, participants' daily usage intention remained high (3.9 ± 0.74), primarily driven by the plaque visualization benefit (4.5 ± 0.85).

Table 3: Toothbrush usability comparison (5-point Likert: 1=Strongly disagree, 5=Strongly agree).

Assessment Item	Score (Mean $\pm$ SD)		
	Oral-B	iPMTC	PIT
Comfortable mouthfeel	4.8 ± 0.42	4.4 ± 0.70	4.7 ± 0.48
Operational convenience	4.7 ± 0.48	3.3 ± 1.06	3.4 ± 0.52
Confidence in plaque removal	3.2 ± 0.92	3.8 ± 0.79	4.5 ± 0.85
Daily use ($\geq 1/\text{day}$)	4.9 ± 0.32	2.6 ± 1.07	3.9 ± 0.74

7 RELATED WORK

In this section, we review the related literature of PIT.

Mobile Oral Health Approaches. The mobile computing community has put forward many oral health methods to improve oral health behaviors and reduce the risk of oral diseases [34]. Approaches to improve oral care habits include promoting oral health knowledge via short messages [35, 36], guiding brushing posture through intelligent mirrors [37], using game-based methods [38–40], and encouraging self-care through social interaction [41]. Feedback on oral care behavior and conditions is also effective. Systems like mORAL [42] provide brushing and flossing feedback, while [43] reminds users to replace their toothbrush. Apps like [44, 45] allow users to take oral photos to detect common diseases. Sensor-based methods assess brushing techniques through brushing coverage and time tracking [2], using technologies such as cameras [39, 46], microphones [47, 48], inertial sensors [49, 50], and photonic sensors [2]. Commercial products like Oral-B iO9 [32] provide feedback on brushing force, and ToMoBrush [51] uses microphones to detect cavities and calculus. However, these systems do not offer direct information about plaque removal during brushing. Prophix [52] uses a camera for oral photos but is positioned too far from the brush head, requiring users to open their mouths widely. iPMTC [14] places the camera closer to the bristles, but it still doesn't help

users align with both visual information and proprioception [15–17]. Additionally, manufacturers recommend using both of these visual toothbrushes only before or after brushing, as the foam often obscures the teeth during brushing [14].

Plaque Detection Methods. Plaque disclosing agents use physical adsorption to bind with dental plaque, making plaque visible to the naked eye. These agents are typically "food-grade dyes" such as basic fuchsin and erythrosine [5]. These agents are considered the gold standard for plaque detection, commonly used by both dentists and families [3]. However, they require an endoscope or mouth opener for clear observation, especially on hard-to-reach inner or outer surfaces of molar [6, 7], and foam can obscure stained plaque during brushing. On the other hand, light-induced fluorescence is another technique for plaque detection. For example, the Qscan device [53] is a portable device that uses blue-violet light to excite plaque metabolites and produce red fluorescence. However, it is limited to detecting only older, mature plaque. Studies have shown that only plaque older than 3 days is likely to be detected [54]. Moreover, other research has found that this technology lacks clinical significance compared to plaque disclosing agents and is considered unreliable for precise plaque quantification [3]. LumiO [6] created a low-fidelity prototype to first explore the integration of plaque detection with a toothbrush. LumiO combines blue-violet LEDs and a camera on the brush head, employing light-induced fluorescence to detect plaque. However, (1) it only detects mature plaque, making it ineffective for identifying and removing new plaque, which is precisely what is required in a brushing scenario. (2) The dual brush head design increases the height, causing discomfort by pressing against the tongue, cheeks, and gums. (3) The rotation on the backside of the brush head may irritate the oral mucosa. (4) It also cannot function with foaming toothpaste, as it fails to counter the foam interference. By contrast, our approach uses a gold-standard plaque disclosing agent to stain plaque. We have also thoughtfully designed the toothbrush head structure for comfort and overcome foam interference, making our solution more practical for real-world use. Note that, PIT targets high-fidelity design, involving a distinct mechanical architecture with key design details to ensure comfort, electrical stability, and production readiness.

8 CONCLUSION AND FUTURE WORK

In this paper, we presented PIT, a novel toothbrush system designed to provide real-time plaque visualization during brushing, overcoming the challenges posed by toothpaste foam interference. PIT carefully designs and deploys toothbrush hardware to provide stable oral visualization at the overlapping areas of the bristles. We also developed a lighting technology to enhance plaque differentiation and designed a deep learning model for precise detection

under thick foam. Finally, we used distilled training to obtain a lightweight student model, which was deployed on a smartphone for evaluation. Through experiments on denture models and human subjects, PIT demonstrates precise and robust plaque segmentation, real-time and low-energy mobile usability, and superior plaque cleaning performance. There are further studies worth exploring and solving in the future.

Interaction Design Optimization: The current visual interaction serves for rapid validation but may overlook user effort optimization. Future work will explore alternative designs (e.g., haptic/auditory plaque cues) to reduce user effort.

Quick Bristles Replacement: Currently, the bristle part cannot be quickly replaced. However, we envision retaining the overall structural design while redesigning the bristle section to a detachable, button-like structure for easier replacement.

Wireless Data Transmission: The connection between the PIT and the smartphone can switch to Bluetooth or Wi-Fi instead of a wired connection to improve mobility.

Training Dataset Expansion: Due to challenges in collecting data from human subjects mentioned in Section 5.1, we employed denture models for training dataset creation. While similar in structure and color to real mouths, denture models lack specific features (e.g., tooth calculus, exposed pulp, or tobacco stains). Therefore, to better approximate real clinical conditions, future work could employ ultrathin customized invisible braces to prevent plaque removal by bristles vibration during data acquisition, thereby ensuring accurate ground-truth alignment.

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